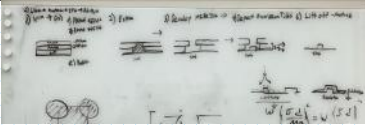


AIScN/SiC IDT Recipe

Monday, March 24, 2025 5:07 PM

- Clean the chip with Hydrofluoric acid (5%) dilute from 10% for 30 seconds
- PMMA file for Alcent from Dalton & talk to him about the fabrication process
- 2/6 post study for Alcent chips in GDS
- Get trained / Fab Chip
- Change GDS to 3.3 GHz
- Fab
- Mag film deposition



UofA Nanofab

RECORD all your steps in classroom notebook AS you do it.

General Solvent Cleaning

1. Sonicate Beakers and tweezers – 10 minutes in DI H₂O (Don't necessarily need to)
2. Sonicate/cook chip in acetone – 5 minutes
3. Sonicate/cook chip in IPA – 5 minutes
4. Sonicate/cook chip in DI H₂O – 5 minutes
5. Blow dry with N₂ gun, create bubble of H₂O on chip and blow off quickly
6. Hotplate bake: 100°C – 10 min. [dehydration step – for most]
7. Optional: O₂ plasma clean, just use 7" or 8" floor O₂ microwave cleaners. 7" floor is lower power than 8" floor. Both are much faster/easier to use than LabTech/Vaportek. Definitely do this for wafers/pipes or other sensitive procedures. Not necessarily needed for depositing electrodes but doesn't hurt.

Notes

It is vital that the chip is very clean before spin coating results. This procedure does a good job of cleaning the chip but will not remove some things occasionally like tape residue. SCE may be worth to clean stickier things. Check if BOE is fine with your chip (Etches SiO₂ and Si₃N₄).

To sonicate or not to sonicate.

AlScN – Sonication is fine

3 um Y cut LN – DO NOT SONICATE, OR DAMP IT WITH ANOTHER BEAKER.

LNCl – CAREFULLY SONICATE/DAMP

Resist spin coating

- CHECK THAT SAMPLE IS CLEAN before spin coating with microscope (if metrics is easiest).
- Use Al foil on outside of spin coater: when done, remove Al and clean inside with acetone or IPA. Whatever dissolves the resist. Acetone for PMMA/PMMA. IPA for discharge.
- Find the correct adapter, bigger ones are easier to set on but can cause more resist to get on the bottom depending on chip size.
- Make sure you are in run mode and NOT edit mode before starting. And vacuum pressure is on

RECORD EBL Wipe off back with acetone needs: Place chip upside down in parabolic container(s) resist isn't touching container, dunk blue swabs in acetone, roll on clean wipers so only a little acetone is left (if you leave too much on there, acetone will get on the front of chip and destroy resist). Or don't do it at all and hope the chip is flat, works most of the time.

Recipes

All current recipes here

EBL/IDT/METAL DEPOSITION RECIPES:

Blayer Depositions typically follow this flow

1. Bottom Layer:
 - Usually a lower molecular weight PMMA (like PMMA 495 A2)
 - More sensitive to electron beam exposure. (lower dose required)
 - Develop faster than the top layer.
 - Thicker than the top layer.
2. Top Layer:
 - Usually a higher molecular weight PMMA (like PMMA 950 A2).
 - Less sensitive to e-beam, (higher dose required)
 - Develops more slowly
 - Thinner than the bottom layer.

Legend: Low Molecular Weight (orange), High Molecular Weight (red), Metal (grey)

Flowchart: Clean Substrate → Spin coat LWR → Spin coat HWR → Spin coat discharge → Pattern with EBL → Develop in MIBK (exposure discharge 1%) → Deposit → LiftOff

EBL IDT Recipe 1) – Thin/High Res – 200nm 495 A2/100nm 950 A4

A) Spin coat

- 1) Follow General Solvent clean and dehydration
- 2) 3kV Beam current – 900 uC/cm² dose – 60k dots – 800 um field size – 0.12 aperture
- 3) REQUIRED TO USE REC
- 4) Develop 60s – 1 day preferably. Can write overnight though.
- 5) Spin – PMMA 950 A2 – 300 rpm – 500 acc. – 2 min (~100nm)
- 6) Spin – PMMA 495 A2 – 300 rpm – 500 acc. – 2 min (~100nm)
- 7) Bake – 180C – 90 sec
- 8) (AlScN – required) Spin Discharge fluid – 1000 rpm – 2 min – right before EBL

9

4 GHz FIDs

3 GHz FIDs

1st test

2nd test

1st test

2nd test

3rd test

4th test

5th test

6th test

7th test

8th test

9th test

10th test

11th test

12th test

13th test

14th test

15th test

16th test

17th test

18th test

19th test

20th test

21st test

22nd test

23rd test

24th test

25th test

26th test

27th test

28th test

29th test

30th test

31st test

32nd test

33rd test

34th test

35th test

36th test

37th test

38th test

39th test

40th test

41st test

42nd test

43rd test

44th test

45th test

46th test

47th test

48th test

49th test

50th test

51st test

52nd test

53rd test

54th test

55th test

56th test

57th test

58th test

59th test

60th test

61st test

62nd test

63rd test

64th test

65th test

66th test

67th test

68th test

69th test

70th test

71st test

72nd test

73rd test

74th test

75th test

76th test

77th test

78th test

79th test

80th test

81st test

82nd test

83rd test

84th test

85th test

86th test

87th test

88th test

89th test

90th test

91st test

92nd test

93rd test

94th test

95th test

96th test

97th test

98th test

99th test

100th test

- Follow EBL Procedure
- 3A Beam current - 900 uA/cm² dose - 60k dots - 600 um field size - 0.12 aperture
- REQUIRED TO USE REC
- Develop 60s ± 1 day after exposure. Can write overnight though.
→ for Alent, use IPA before MIBK to remove discharge.
- Develop (ASAR - 1 day after exposure)
(DI H₂O is a strong developer, so must not have any in beaker)
- MIBK - 1-15 min. (You can lightly stir, but be consistent)
- Quench in IPA - 15sec - 1.00 min (Dose stop development)
- If still underdeveloped do a 5-10 more seconds until done.
- OK do codeveloper DI H₂O IPA 1:3 for 5 seconds will do the trick. But careful its very fast.

D) Deposit Metal

- Follow Thermal Procedure. Use a recipe in a folder, visible to all.
- LiR Off
- Acetone Overnight. Tilt chip if you can. Minimum probably ~2 hours. Tell it comes off. Can heat acetone too if it's a rush. DO NOT BOIL - or other developer.
- Grab a needle and shoot off metal gently. Or swirl around. Or can do it overnight upside down. Hard to do though. Or on its side is
- If there's metal in between fingers. Do a quick sonication demand. Should come right off.
→ Use N₂ for lift-off with help plate and reduce sonication and add second layer of water bath when sonication.
- Notes:
If didn't work, its usually one of 3 things. Dose was off. Develop was off.

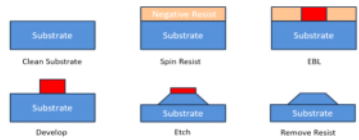
EBL IOT Recipe 21 - Medium/Medium Res - 200nm EL 11/100nm 950 A4

EBL IOT Recipe 21 - Thick/Low Res - 200nm EL 11/100nm 950 A4

MLA IOT/METAL DEPOSITION RECIPES:

ETCHING RECIPES:

General Negative Etch Recipe typically follows this flow



Lithium Niobate Dry Etching - Argon

ALSCN Wet Etching:

Chip cleaning

For Alent/Sin

① BoE 10:1

↳ 1-20 minutes in 30.4 parts (3 times)

② Use plasma clamp

③ Check metal & work

④ Discard 30.4 in waste ⑤ R&D goes in waste

⑥ Sonicate Alent IPA (30s each)

⑦ Plasma clean (5 min)

⑧ Check using microscope.

9

[X] 0.34

[X] 0.32

[X] 0.3

[X] 0.28

[X] 0.26

[X] 0.24

[X] 0.34

[X] 0.32

[X] 0.3

[X] 0.28

[X] 0.26

[X] 0.34

[X] 0.32

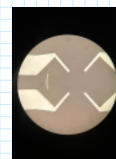
[X] 0.3

[X] 0.28

[X] 0.26



0.36



0.46

Notes

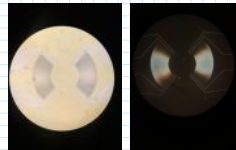
- N₂ too much air pressure
- Not Removing Discharge before developing
- PEC files Linbo₃ → Alent
- Too much Sonication

PEC → LN

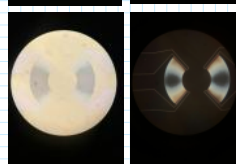
1:00 MIBK

PEC → ALSCN

1:15 MIBK + 3 sec DI H₂O: IPA



lowest dose
0.36

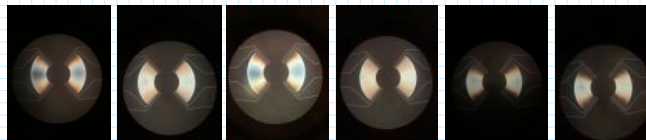


highest dose
0.46

Light field
50x

Dark field
50x

Dose test vs GHz FIDT → Dark field 50x After developing



0.36

0.38

0.40

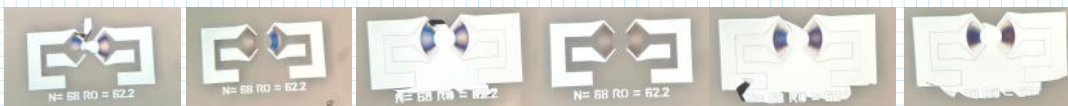
0.42

0.44

0.46



0.40 0.36



N= 68 RD = 62.2

N= 68 RD = 62.2

N= 68 RD = 62.2

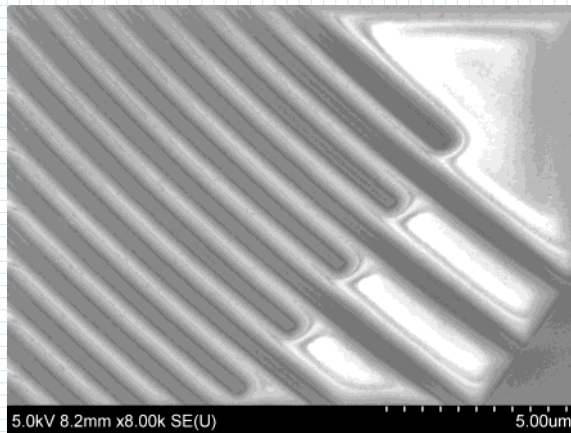
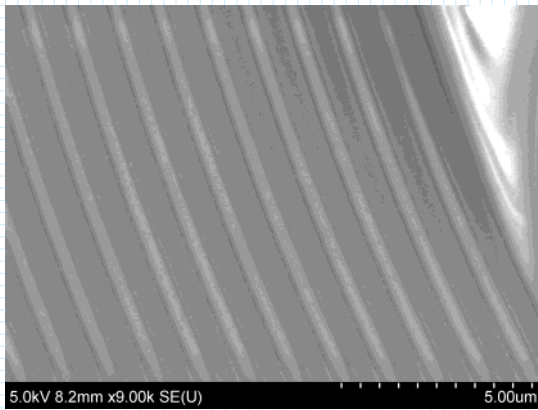
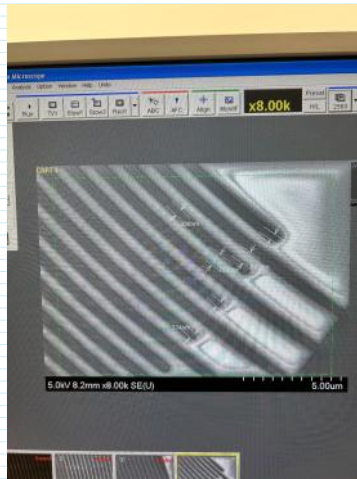
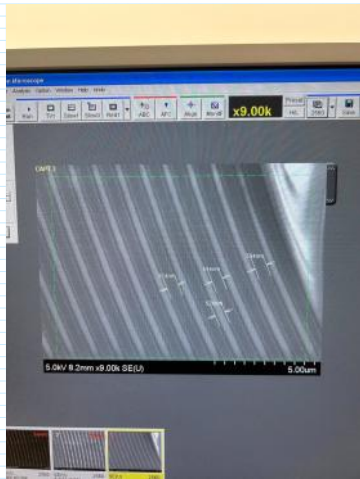
N= 68 RD = 62.2

N= 68 RD = 62.2

N= 68 RD = 62.2

Lowest → Highest



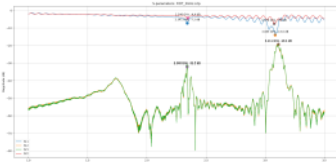
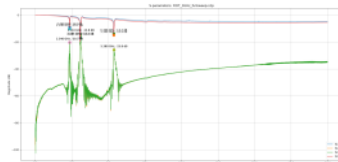
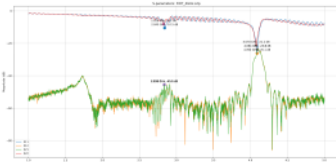


Terminal
Reimage
for Gold Cap
+ MLA (Pads)

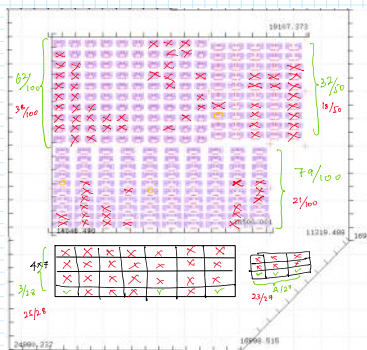
Ti 10nm
Al 10nm
Au 5nm

AZ 501A → Photoresist
AZ 100K → Developer
Microdev
MIF 319 ✓

Check the
chemicals
in MLA room



$\lambda = 2.52 \mu m$
 $f = 3 \text{ GHz}$ (Suzawa mode)
 $f = 2.5 \text{ GHz}$ (Rayleigh mode)
 $f = 5.3 \text{ GHz}$ (Higher Harmonic?)



What went wrong?

- FBL
 - 1) Beam drift
 - 2) Sample inclination
 - 3) Dose rate drift

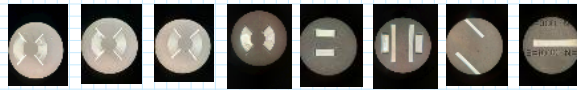
dot Devices
150 survived
125 failed
40% loss
60% survival

After Lift-off Layer 1 only:

FOTs
66/100
Survived.



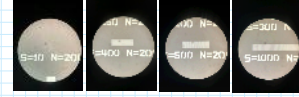
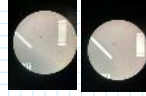
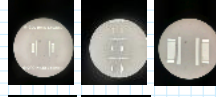
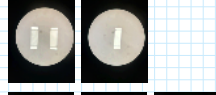
20/50
Survived.



IDTs
80/100
Survived.

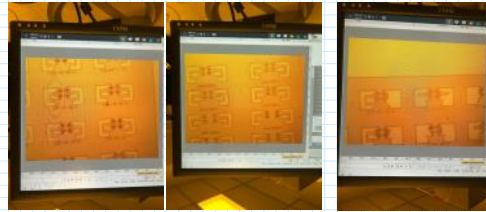
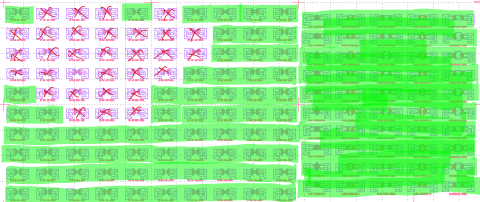


Resonators
23/24
Survived.

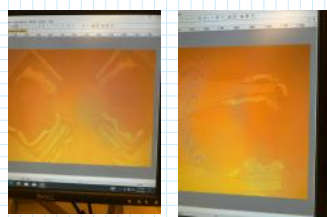
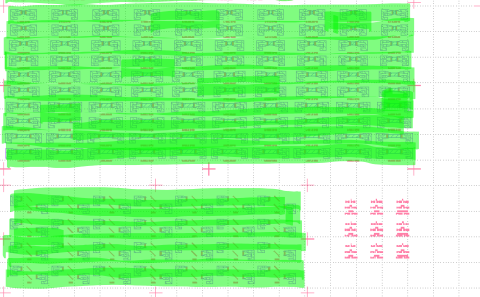


Gratings
3/27
Survived.

After MLA
develop.



overexposed
AC + C Acetone
in MFE 319
before washing
- delayed



Metal Deposition : Ti = 10nm, Al = 190 nm for GSG pads



